

HOSPITAL OPERATIONS & PATIENT ANALYTICS DASHBOARD

(STORYBOARD)

Infosys Springboard Internship Project

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1. PROJECT OVERVIEW

Project Overview

Healthcare organizations generate large volumes of operational data every day through patient admissions, room allocations, bed management, medical departments, and healthcare resources. However, raw data alone does not provide meaningful insights unless it is analyzed and presented in a structured manner. Hospital administrators often require a centralized view of hospital operations to monitor performance, evaluate resource utilization, and make timely decisions that improve patient care and operational efficiency.

The **MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)** project was developed to transform hospital operational data into an interactive and easy-to-understand business intelligence solution using **Power BI**. The project integrates data from multiple hospital-related datasets, including patient records, admissions, departments, rooms, beds, doctors, nurses, and medical equipment, to provide a comprehensive overview of hospital performance.

Instead of relying on multiple reports, the dashboard suite brings together key operational metrics and visualizations into four interconnected dashboards: **Hospital Overview**, **Patient Flow**, **Department Analytics**, and **Resource Utilization**. Each dashboard focuses on a specific aspect of hospital operations while remaining connected through interactive filters and navigation, enabling users to explore data from different perspectives.

The primary goal of this project is to help hospital management understand operational performance, identify trends, compare departmental activities, monitor resource utilization, and support data-driven decision-making. By presenting healthcare information through meaningful KPIs and interactive visualizations, the dashboard makes complex hospital data more accessible and actionable for decision-makers.

Key Technologies Used

- **Power BI** – Dashboard development and interactive visualizations
- **Python (Pandas & NumPy)** – Data cleaning, preprocessing, and KPI engineering
- **Microsoft Excel** – Intermediate data preparation
- **DAX (Data Analysis Expressions)** – KPI calculations and business measures

Project Outcome

The final solution consists of a **four-dashboard interactive Power BI report** that enables users to monitor hospital performance, analyze patient flow, evaluate departmental operations, and optimize healthcare resource utilization through a single integrated analytics platform.

2. PROJECT OBJECTIVES

Project Objectives

The primary objective of this project is to develop an interactive hospital analytics dashboard that provides a clear and centralized view of hospital operations. By transforming raw healthcare data into meaningful visualizations and key performance indicators (KPIs), the dashboard enables users to monitor hospital performance, analyze patient flow, evaluate departmental operations, and track healthcare resource utilization through an intuitive interface.

To achieve this objective, the project focuses on the following goals:

- Collect, clean, and integrate hospital operational datasets into a unified data model.
- Develop meaningful healthcare KPIs such as **Total Admissions, Occupancy Rate, Average Length of Stay (LOS), Readmission Rate, Bed Utilization Rate, and Department Efficiency**.
- Design four interconnected dashboards that provide insights into different aspects of hospital operations:
 - Hospital Overview
 - Patient Flow
 - Department Analytics
 - Resource Utilization
- Enable users to analyze hospital data through interactive filters and cross-dashboard navigation.
- Support hospital administrators and management teams in identifying operational trends, monitoring departmental performance, improving patient care, and optimizing resource allocation.

Expected Outcomes

By the end of the project, the dashboard is expected to:

- Provide a comprehensive overview of hospital performance through real-time KPIs and visualizations.
- Help users understand patient admission, discharge, and readmission patterns.
- Compare departmental performance based on staffing, patient volume, efficiency, and treatment capacity.
- Monitor the utilization and availability of beds, rooms, staff, and medical equipment.
- Assist decision-makers in planning resources more effectively and making informed operational decisions using data-driven insights.

3. TARGET USERS

Target Users

The MedTrack_DV dashboard is designed to support different stakeholders involved in hospital operations. Since each user has different responsibilities, the dashboard provides multiple views that help them access relevant information quickly and make informed decisions based on real-time operational insights.

Hospital Administrators

Hospital administrators require a high-level overview of the hospital's overall performance. They use the dashboard to monitor key performance indicators such as total admissions, occupancy rate, average length of stay, revenue, and bed utilization. These insights help them evaluate hospital performance, identify operational trends, and support strategic planning.

Healthcare Managers

Healthcare managers focus on the day-to-day functioning of hospital operations. The Patient Flow dashboard helps them monitor patient admissions, discharges, readmissions, and patient demographics. By analyzing patient movement and average length of stay, they can identify bottlenecks, improve patient flow, and enhance service delivery.

Department Heads

Department heads are responsible for managing resources within their respective departments. The Department Analytics dashboard enables them to compare staffing levels, patient volume, treatment capacity, equipment availability, and departmental efficiency. These insights help them identify workload differences between departments and support better resource planning.

Hospital Operations Team

The operations team uses the Resource Utilization dashboard to monitor the availability and utilization of hospital resources, including beds, rooms, staff, and medical equipment. Tracking equipment usage, maintenance status, bed occupancy, and staff allocation helps ensure that resources are efficiently utilized while maintaining smooth hospital operations.

Benefits to Stakeholders

The dashboard provides different levels of insight depending on the user's role while maintaining a consistent and interactive reporting experience. By integrating operational data into a single analytics platform, it enables stakeholders to:

- Monitor hospital performance through interactive KPIs and visualizations.
- Analyze patient flow and identify operational bottlenecks.
- Compare departmental performance and treatment capacity.
- Track healthcare resource utilization and availability.
- Support faster, data-driven operational and strategic decision-making.

4. DASHBOARD DESIGN STRATEGY

Dashboard Design Strategy

The dashboard was designed to provide a structured and user-friendly view of hospital operations by organizing information into four interconnected dashboards. Instead of displaying all metrics on a single page, the information is divided into separate dashboards based on different operational areas. This approach improves readability, reduces visual clutter, and enables users to focus on specific aspects of hospital performance while maintaining a consistent navigation experience.

Each dashboard serves a distinct purpose within the overall analytics solution:

- **Hospital Overview** provides a summary of the hospital's overall operational performance through key performance indicators and monthly trends.
- **Patient Flow** focuses on patient admissions, discharges, readmissions, demographics, and movement across the hospital.
- **Department Analytics** compares departmental performance by analyzing staffing, patient volume, department efficiency, equipment, and treatment capacity.
- **Resource Utilization** monitors the utilization and availability of hospital resources, including beds, rooms, staff, and medical equipment.

To provide a seamless user experience, a navigation menu is placed at the top of every dashboard, allowing users to switch easily between the four analytical views. Each dashboard also includes interactive slicers that allow users to filter the data based on the most relevant dimensions for that page, enabling more detailed analysis without affecting the overall layout.

The dashboards follow a consistent visual design using a dark theme with contrasting colors to improve readability and highlight important information. KPI cards are positioned at the top of each dashboard to provide an immediate summary of key metrics, while charts are arranged below them to present supporting trends, comparisons, and operational insights. This consistent layout allows users to quickly interpret information and navigate between dashboards with minimal effort.

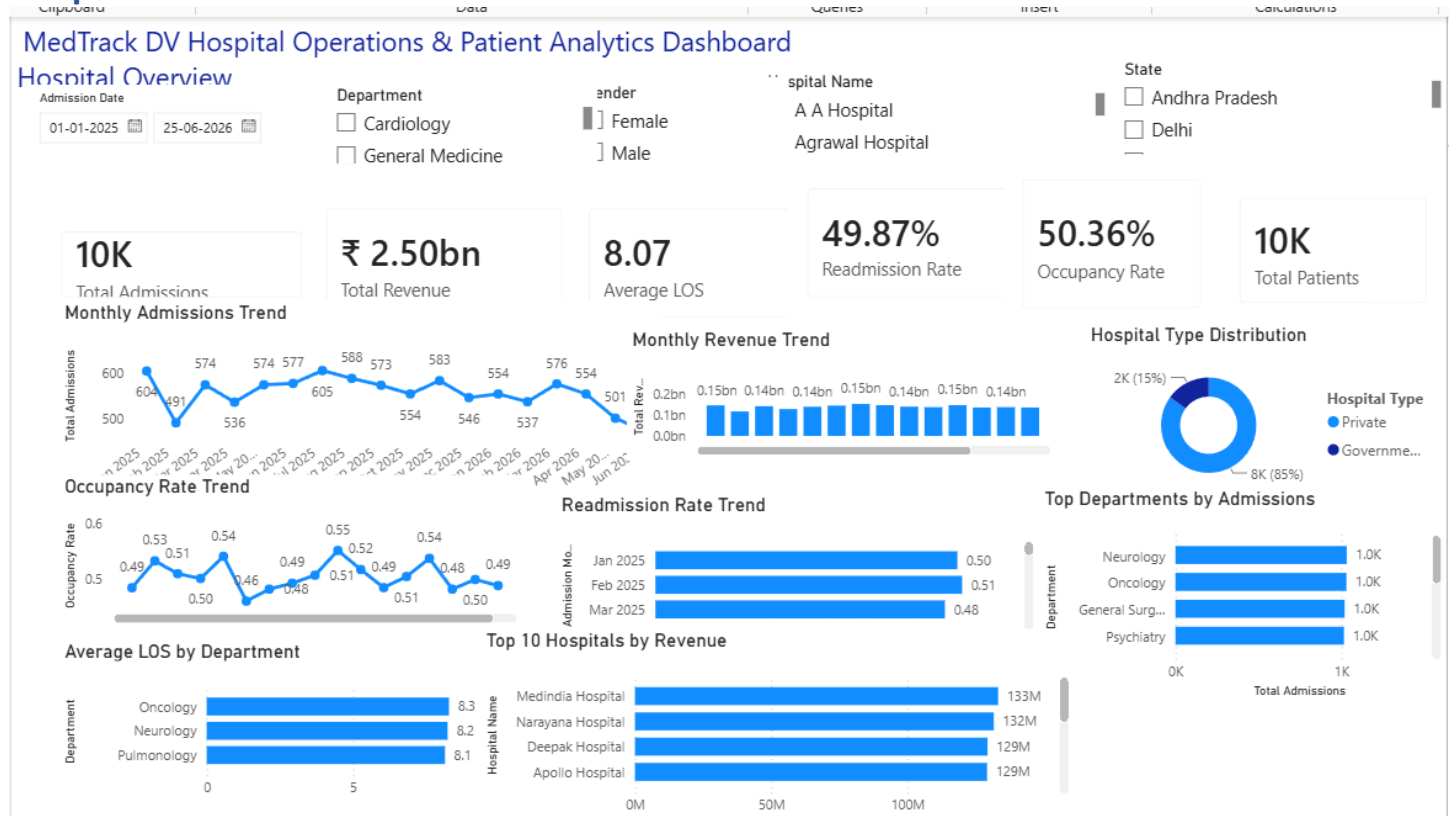
The overall design emphasizes clarity, consistency, and interactivity, ensuring that hospital administrators and healthcare managers can efficiently monitor hospital operations, compare performance across departments, and make informed decisions using a centralized business intelligence solution.

Design Principles Followed

- Simple and consistent dashboard layout across all pages.
- Clear separation of executive, operational, departmental, and resource-related analytics.
- Interactive slicers to enable flexible data exploration.
- Uniform color palette, typography, and KPI placement for consistency.
- Business-focused visualizations that support quick understanding and decision-making.
- Responsive dashboard navigation for a smooth user experience.

5. DASHBOARD 1

Hospital Overview



Purpose

The **Hospital Overview** dashboard provides a high-level summary of the hospital's operational performance. It is designed for hospital administrators and senior management who require a quick understanding of the hospital's overall performance without analyzing detailed operational data. The dashboard consolidates key performance indicators and monthly trends into a single view, allowing decision-makers to monitor hospital efficiency and identify areas that require attention.

Business Questions Addressed

This dashboard was designed to answer the following business questions:

- How many patients are being served by the hospital?
- What is the total number of admissions during the selected period?
- How efficiently are hospital beds being utilized?
- What is the average length of stay for admitted patients?
- How is the hospital performing in terms of revenue generation?
- Are readmission rates showing any concerning trends?
- Which departments contribute the highest revenue?
- How does occupancy change over time?

Key Performance Indicators (KPIs)

The dashboard highlights six key operational indicators that provide an executive summary of hospital performance:

- Total Patients
- Total Admissions
- Total Revenue
- Average Length of Stay (LOS)
- Bed Utilization Rate
- Occupancy Rate

These KPIs provide a quick overview of hospital operations and help management monitor overall performance without examining detailed reports.

Dashboard Visualizations

The dashboard includes several interactive visualizations that support the KPIs and provide additional operational insights:

- Monthly Admissions Trend to monitor changes in patient admissions over time.
- Revenue Trend to analyze hospital revenue across different months.
- Readmission Trend to identify changes in patient readmissions.
- Occupancy Trend to monitor hospital occupancy patterns.
- Appointment Status Distribution to understand appointment outcomes.
- Top Revenue Departments to identify departments contributing the highest revenue.
- Average Length of Stay by Department to compare patient stay duration across departments.

Each visualization was selected to complement the KPIs and provide both trend analysis and departmental comparisons.

Business Value

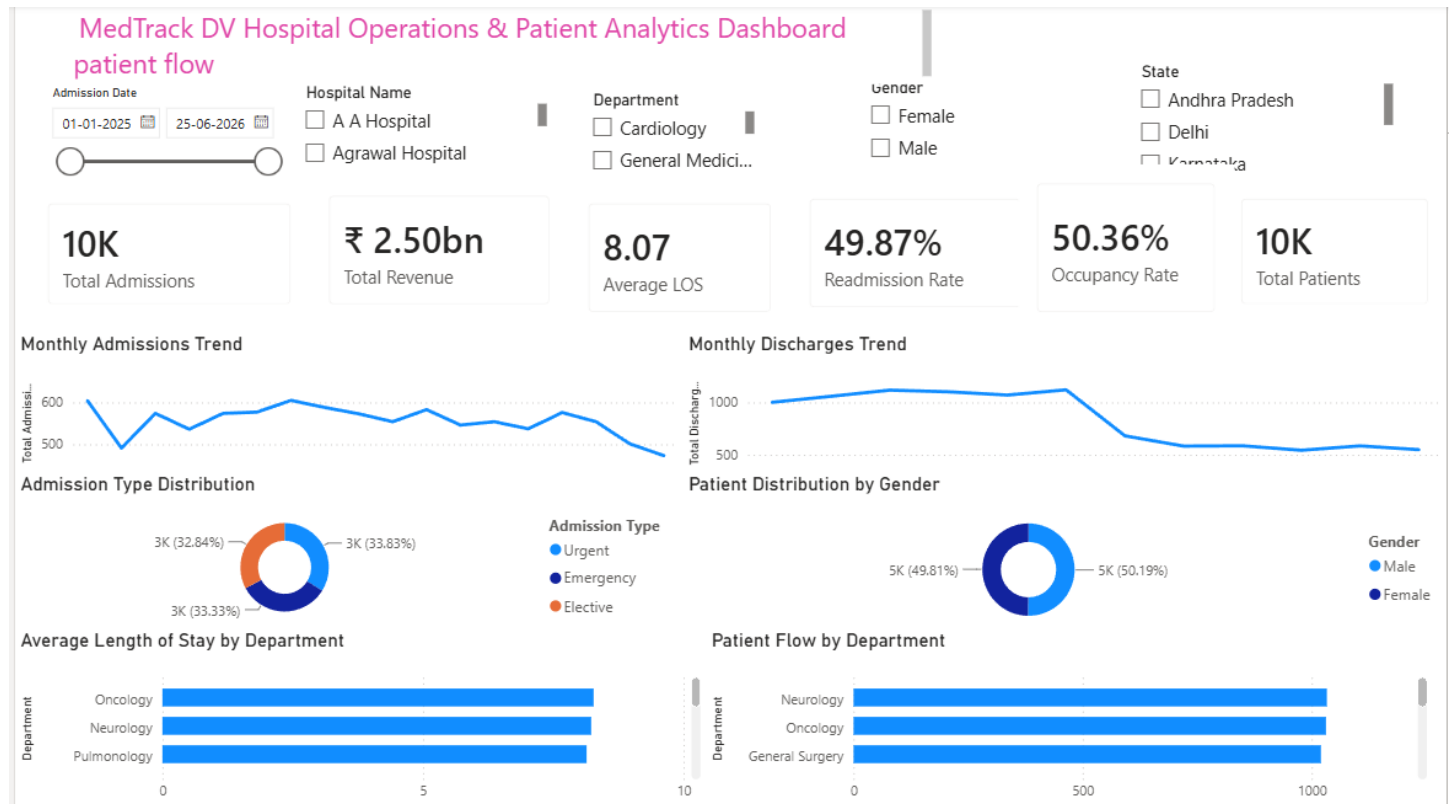
The Hospital Overview dashboard serves as the executive summary of the entire reporting solution. It enables hospital management to evaluate operational performance from a single interface, identify emerging trends, monitor resource utilization, and compare departmental performance. By presenting critical healthcare metrics in an interactive format, the dashboard supports faster and more informed decision-making while reducing the effort required to analyze operational reports.

Key Insight

This dashboard provides a consolidated view of the hospital's overall operational health by combining patient, financial, and resource-related metrics. The integration of KPIs with trend analysis enables decision-makers to quickly identify performance changes, monitor hospital efficiency, and prioritize areas that may require operational improvements.

6. DASHBOARD 2

Patient Flow



Purpose

The **Patient Flow** dashboard focuses on analyzing how patients move through the hospital, from admission to discharge, while also monitoring readmissions and patient demographics. It provides healthcare managers and hospital administrators with insights into patient movement, treatment duration, and readmission patterns. By presenting these operational metrics in an interactive format, the dashboard helps identify bottlenecks, improve patient management, and support better planning of hospital services.

Business Questions Addressed

This dashboard is designed to answer the following business questions:

- How many patients were admitted and discharged?
- What is the average length of stay for admitted patients?
- How many patients were readmitted?
- Which departments handle the highest number of admissions and discharges?
- How are patients distributed across different age groups and gender?
- Are there departments with higher readmission levels that require attention?

Key Performance Indicators (KPIs)

The dashboard summarizes patient flow using six operational KPIs:

- Unique Patients
- Total Admissions
- Total Discharges
- Average Length of Stay (LOS)
- Total Readmissions
- Readmission Rate

These KPIs provide a quick overview of patient movement and help evaluate the efficiency of hospital operations.

Dashboard Visualizations

The dashboard includes interactive visualizations that provide detailed insights into patient flow:

- **Admissions by Department** to identify departments with the highest patient intake.
- **Discharges by Department** to compare discharge volumes across departments.
- **Admissions vs. Discharges by Department** to evaluate patient movement and departmental workload.
- **Patient Distribution by Gender** to understand demographic distribution.
- **Patient Distribution by Age Group** to analyze patient demographics.
- **Readmissions by Department** to identify departments with higher readmission counts.

Each visualization supports a different aspect of patient flow analysis and complements the KPIs displayed at the top of the dashboard.

Business Value

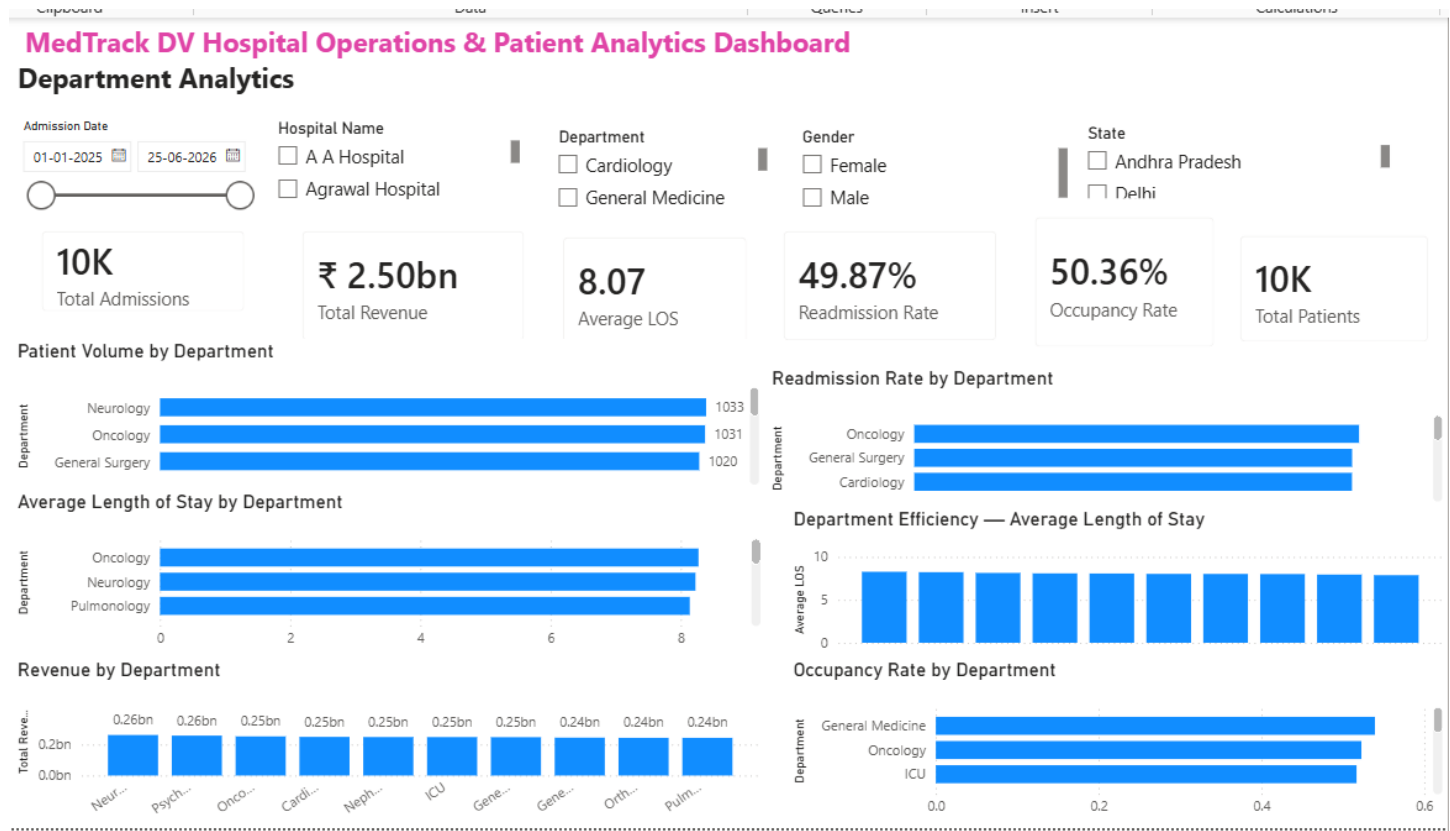
The Patient Flow dashboard enables hospital management to monitor the complete patient journey and evaluate the efficiency of admission and discharge processes. It helps identify departments experiencing higher patient volumes or readmission levels and supports decisions related to patient care, discharge planning, and operational improvements. The demographic analysis also provides additional context for understanding patient populations and planning healthcare services.

Key Insight

The Patient Flow dashboard provides a comprehensive view of hospital patient movement by combining operational KPIs with demographic and departmental analysis. It helps decision-makers monitor admission and discharge trends, evaluate readmission patterns, and understand patient demographics, enabling more efficient patient management and improved healthcare service delivery.

7. DASHBOARD 3

Department Analytics



Purpose

The **Department Analytics** dashboard is designed to evaluate and compare the operational performance of different hospital departments. It provides insights into staffing, patient volume, departmental efficiency, equipment availability, and treatment capacity, enabling hospital management to identify workload distribution and resource requirements across departments.

Rather than focusing on overall hospital performance, this dashboard helps users understand how individual departments are performing and whether available resources are aligned with patient demand.

Business Questions Addressed

This dashboard is designed to answer the following business questions:

- Which departments handle the highest patient volume?
- How are doctors and nurses distributed across different departments?
- Which departments have better operational efficiency?
- Are medical equipment and rooms distributed adequately among departments?
- Which departments have greater treatment capacity?

Key Performance Indicators (KPIs)

The dashboard summarizes departmental performance using the following KPIs:

- Total Departments
- Total Doctors
- Total Nurses
- Total Equipment
- Total Equipment Usage Hours
- Department Efficiency (Patients per Doctor)

These KPIs provide a quick overview of departmental resources and operational efficiency.

Dashboard Visualizations

The dashboard includes the following visualizations to compare departmental performance:

- **Doctors by Department** to compare doctor allocation across departments.
- **Nurses by Department** to analyze nursing staff distribution.
- **Patient Volume by Department** to identify departments serving the highest number of patients.
- **Equipment Inventory by Department** to evaluate equipment availability.
- **Department Efficiency (Patients per Doctor)** to compare operational efficiency across departments.
- **Rooms by Department** to assess treatment capacity and infrastructure availability.

Each visualization focuses on a different aspect of departmental performance, allowing users to compare resources, workload, and efficiency from multiple perspectives.

Business Value

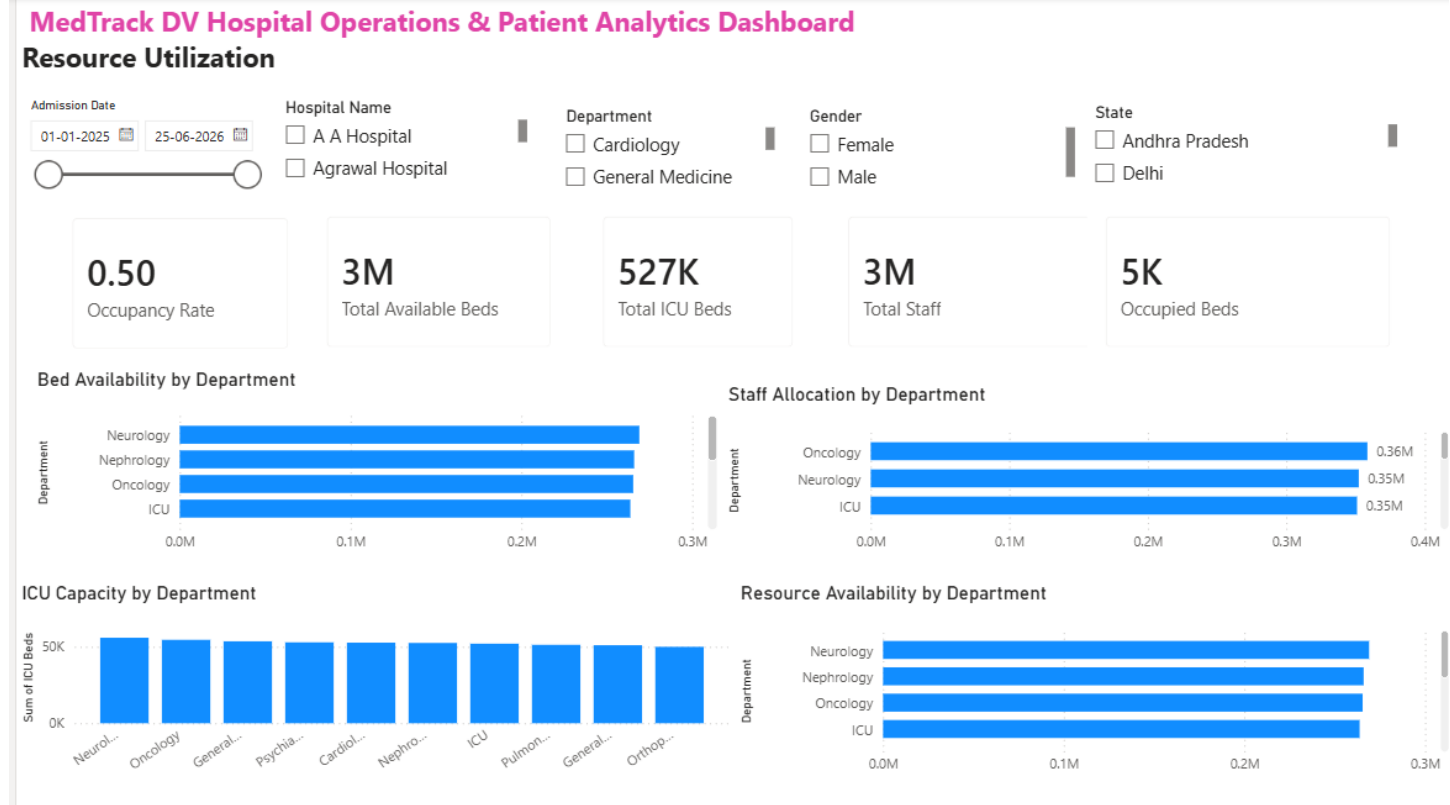
The Department Analytics dashboard enables hospital administrators and department heads to compare operational performance across departments using staffing, patient volume, efficiency, and infrastructure metrics. These insights support better workforce planning, balanced resource allocation, and informed decisions regarding departmental capacity and performance improvement.

Key Insight

By combining patient volume, staffing levels, departmental efficiency, equipment inventory, and room availability into a single analytical view, this dashboard provides a comprehensive comparison of departmental operations. It helps identify departments with higher workloads, evaluate whether resources are allocated appropriately, and support data-driven decisions that improve operational efficiency and patient care.

8. DASHBOARD 4

Resource Utilization



Purpose

The **Resource Utilization** dashboard is designed to monitor how effectively the hospital utilizes its physical and operational resources. It provides a comprehensive view of resource availability, staff allocation, bed utilization, and equipment usage, helping hospital administrators optimize resource planning and ensure that healthcare services are delivered efficiently.

The dashboard focuses on maximizing the utilization of available resources while maintaining operational readiness. By monitoring the status and usage of hospital assets, decision-makers can identify underutilized or overutilized resources and take corrective actions to improve overall hospital efficiency.

Business Questions Addressed

This dashboard is designed to answer the following business questions:

- How effectively are hospital beds being utilized?
- How are healthcare staff allocated across different departments?
- What is the current availability of medical equipment?
- Which equipment types are most commonly used?
- How does equipment usage change over time?

- Are hospital resources sufficient to support operational demands?

Key Performance Indicators (KPIs)

The dashboard summarizes resource utilization using the following KPIs:

- Total Beds
- Total Wards
- Total Rooms
- Bed Utilization Rate
- Average Equipment Usage Days
- Equipment Availability Rate

These KPIs provide an overall view of the hospital's resource capacity, utilization, and availability.

Dashboard Visualizations

The dashboard includes several visualizations that support operational resource management:

- **Bed Occupancy by Ward** to compare bed utilization across different wards.
- **Staff Allocation by Department** to analyze the distribution of doctors, nurses, and helpers among departments.
- **Equipment Type Distribution** to understand the availability of different categories of medical equipment.
- **Equipment Status Distribution** to monitor equipment that is currently available, in use, or under maintenance.
- **Maintenance Flag Distribution** to identify equipment requiring maintenance attention.
- **Monthly Equipment Usage Hours** to analyze equipment utilization trends over time.

Each visualization provides operational insights that help hospital management evaluate resource allocation, identify utilization patterns, and improve capacity planning.

Business Value

The Resource Utilization dashboard supports efficient management of hospital resources by providing visibility into bed occupancy, staff allocation, equipment utilization, and resource availability. These insights help administrators optimize resource distribution, improve operational planning, reduce equipment downtime, and ensure that hospital facilities are adequately prepared to meet patient demand.

Key Insight

This dashboard brings together infrastructure, workforce, and equipment analytics into a single operational view. By monitoring resource utilization, availability, and maintenance status, hospital administrators can make informed decisions regarding capacity planning, workforce allocation, and equipment management. The interactive design enables users to quickly identify resource constraints and opportunities for improving operational efficiency.

9. FILTERS & INTERACTIVITY

Filters & Interactive Features

To enhance usability and improve data exploration, the dashboard suite incorporates interactive filters, navigation controls, and cross-filtering capabilities. These features allow users to analyze hospital data from different perspectives without modifying the underlying dataset. Instead of displaying static reports, the dashboards provide an interactive experience where users can focus on specific departments, wards, patient groups, or operational resources based on their analytical requirements.

Each dashboard includes filters that are specifically designed for its analytical purpose, ensuring that users can quickly access relevant information while maintaining a clean and organized layout.

Dashboard Filters

The filters implemented across the dashboard suite are:

- **Hospital Overview**
 - Date Range
 - Department
 - Gender
- **Patient Flow**
 - Date Range
 - Department
 - Gender
- **Department Analytics**
 - Department
 - Equipment Type
 - Equipment Name
- **Resource Utilization**
 - Ward
 - Department
 - Equipment Name

These filters allow users to focus on specific operational areas while ensuring that the displayed KPIs and visualizations update dynamically based on the selected criteria.

Cross-Filtering

The dashboard also supports cross-filtering, where selecting a value in one visualization automatically updates the related visualizations on the same page. This enables users to perform interactive analysis and quickly identify patterns without creating additional reports.

For example, selecting a particular department immediately updates the related KPIs and charts to display information only for the selected department. Similarly, selecting a ward or equipment category provides a focused view of resource utilization within that specific area.

Navigation

To provide a seamless reporting experience, navigation buttons are placed at the bottom of each dashboard. These buttons allow users to move easily between the four dashboards:

- Hospital Overview
- Patient Flow
- Department Analytics
- Resource Utilization

This consistent navigation structure enables users to transition between executive summaries, patient analytics, departmental comparisons, and resource monitoring without leaving the report.

Business Value

The interactive design significantly improves the usability of the dashboard by allowing users to explore healthcare data from multiple perspectives. Instead of relying on static reports, hospital administrators can apply filters, compare departments, monitor patient flow, and evaluate resource utilization in real time. This flexibility supports faster analysis, improves decision-making, and provides a more comprehensive understanding of hospital operations.

10. KEY BUSINESS INSIGHTS

Key Business Insights

The dashboard provides meaningful insights into different aspects of hospital operations by combining patient, departmental, financial, and resource-related data into a single business intelligence solution. These insights enable hospital administrators to identify trends, monitor operational performance, and make informed decisions to improve efficiency and patient care.

Some of the key business insights generated from the dashboard include:

1. Hospital Performance Monitoring

The Hospital Overview dashboard provides a consolidated view of the hospital's operational performance through KPIs such as Total Admissions, Total Patients, Revenue, Occupancy Rate, Bed Utilization Rate, and Average Length of Stay. These indicators help management quickly assess the overall health of hospital operations.

2. Patient Flow Analysis

The Patient Flow dashboard highlights admission, discharge, and readmission patterns, allowing management to monitor patient movement throughout the hospital. Patient demographic analysis by age group and gender also provides a better understanding of the hospital's patient population and supports effective healthcare planning.

3. Departmental Performance Comparison

The Department Analytics dashboard enables comparison of different hospital departments based on staffing levels, patient volume, equipment availability, treatment capacity, and department efficiency. This comparison helps identify departments with higher workloads and supports better resource allocation.

4. Resource Utilization Monitoring

The Resource Utilization dashboard provides visibility into the availability and utilization of hospital resources, including beds, rooms, staff, and medical equipment. Equipment status, maintenance requirements, and utilization trends help ensure that hospital resources are effectively managed and available when required.

5. Capacity Planning Support

The combination of bed occupancy, room availability, staff allocation, and equipment utilization enables hospital administrators to evaluate current operational capacity and identify areas where additional resources or improved planning may be required.

6. Interactive Decision Support

The use of interactive filters, cross-filtering, and dashboard navigation allows users to analyze hospital operations from multiple perspectives. This flexibility enables faster investigation of operational issues and supports data-driven decision-making without the need for multiple static reports.

Overall Business Impact

By integrating operational, departmental, patient, and resource-related information into a single interactive reporting solution, the dashboard transforms raw hospital data into meaningful business insights. It enables healthcare stakeholders to monitor performance, optimize resource utilization, improve departmental planning, and make informed decisions that contribute to more efficient hospital operations and enhanced patient care.

11. PROJECT OUTCOMES

Project Outcomes

The MedTrack_DV dashboard successfully transformed hospital operational data into an interactive business intelligence solution that provides meaningful insights into hospital performance, patient management, departmental operations, and resource utilization. By integrating multiple hospital datasets into a unified Power BI report, the project enables users to analyze complex healthcare data through intuitive visualizations and key performance indicators.

The project achieved the following outcomes:

- Developed a centralized reporting solution consisting of four interconnected dashboards, each focusing on a specific aspect of hospital operations.
- Converted raw hospital data into meaningful KPIs and interactive visualizations that support operational monitoring and performance evaluation.
- Enabled users to analyze patient flow, departmental performance, and resource utilization through dynamic filters and cross-filtering capabilities.
- Provided comparative analysis across hospital departments to support workforce planning, treatment capacity evaluation, and operational efficiency.
- Improved visibility into hospital resources by monitoring bed utilization, equipment availability, staff allocation, and maintenance activities.
- Created an intuitive dashboard design that allows hospital administrators and healthcare managers to access critical information quickly and make informed decisions.

Project Benefits

The completed dashboard offers several practical benefits for healthcare organizations:

- Provides a centralized platform for monitoring hospital operations.
- Reduces the time required to analyze operational data through interactive dashboards.
- Supports better planning of beds, rooms, staff, and medical equipment.
- Helps identify operational trends and departmental workload differences.
- Enhances decision-making using dynamic KPIs and interactive visualizations.
- Improves communication of operational performance through clear and business-oriented reporting.

Overall Outcome

The MedTrack_DV dashboard demonstrates how business intelligence can transform raw healthcare data into actionable insights. By combining interactive dashboards, meaningful KPIs, and operational analytics, the project supports hospital administrators in monitoring performance, optimizing resource utilization, improving departmental operations, and making informed, data-driven decisions.

12. CONCLUSION

Conclusion

The **MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)** project successfully demonstrates how business intelligence can be used to transform hospital operational data into meaningful and actionable insights. By integrating data from multiple hospital departments and presenting it through interactive Power BI dashboards, the project provides a comprehensive view of hospital performance, patient flow, departmental operations, and resource utilization.

The four dashboards work together to provide both executive-level and operational insights. Hospital administrators can monitor overall performance through key performance indicators, healthcare managers can analyze patient movement and readmission trends, department heads can evaluate staffing and operational efficiency, and the operations team can monitor the utilization and availability of hospital resources.

Throughout this project, emphasis was placed on developing a solution that is not only visually appealing but also practical and business-oriented. Interactive filters, meaningful KPIs, consistent dashboard design, and intuitive navigation enable users to explore data efficiently and make informed decisions based on real-time operational insights.

Overall, this project highlights the importance of data-driven decision-making in healthcare. By converting raw hospital data into an interactive analytics platform, the dashboard supports better operational planning, efficient resource utilization, improved departmental performance, and enhanced patient care. It demonstrates the practical application of **Power BI, Python, DAX, and data analytics techniques** in solving real-world healthcare management challenges.

Final Reflection

Developing this project provided valuable hands-on experience in data preprocessing, data modeling, KPI development, dashboard design, and business analytics. More importantly, it strengthened the understanding of how analytical solutions can support healthcare organizations in making informed decisions and improving overall operational efficiency. The MedTrack_DV dashboard serves as a practical example of applying business intelligence to address real-world healthcare challenges through interactive and insightful data visualization.